

Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	First	Academic Year	2024/2025

Course Details

Course Title	Operating Systems			Course No.		A0413301 (0433301)	
Type of Learning	☒ On- campus ☐ Blended (Platform) ☐ Online (Platform)						
Credit Hours	3	Theoretical	2	Practical	1		
Course Level (according to JNQF)			7	JNQF Hours*		12.1	
Pre-requisite	A0412101 (0432101) - Data Structures			Co-requisite		N/A	

Course Instructor Information

Instructor Name	Dr. Rashed Alazzeh		
Instructor E-mail	razzeh@meu.edu.jo		
Course Time			
Office Hours			
Office Number	N 108	Office Phone	
Lab Instructor Name –If assigned-	Dr. Rashed Alazzeh		

Sources and References

1. Textbook

Modern Operating Systems, Global Edition Paperback – April 24, 2023
by Andrew Tanenbaum (Author)

2. References

- Operating Systems: Internals and Design Principles, 9th Edition. Willian Stallings. 2018. Wiley
- Operating System Concepts 10th Edition
by Abraham Silberschatz (Author), Peter B. Galvin (Author), Greg Gagne (Author)

Teaching Methods

- | | | |
|----------------------|-----------------------------------|---------------------------|
| 1. Lectures | 2. Discussion and problem solving | 3. Individual assignments |
| 4. Groups discussion | 5. Laboratories | |

Course Description

The core concepts of operating systems, such as: Process: Critical Sections; Mutual exclusion; Semaphores; Monitors. Classical Problems: Deadlock; Process Scheduling. Memory Management: Multi-Programming; Swapping; Virtual Memory; Paging and Segmentation. File System: Operations; Implementation; Performance.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Compare the functionality of different computing hardware structures and Operating Systems Structures.	PLO 1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	X		
2	Discuss issues of Process Management including process structure, synchronization, scheduling, deadlock and communication.	PLO2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		X	
3	Demonstrate memory management issues including advance techniques of paging, segmentation and virtual memory.	PLO 1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	X		
4	Explain the operation of various File Management Algorithms.	PLO2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		X	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	Course Syllabus, Introduction; Computer system organization and architecture, Theoretical Lab (Introduction)	Lecture (1)	Face to face	1		
		Lecture (2)	Face to face	2		
		Lab (1)	Lab	2		

2	Operating systems structure and Operations, process management	Lecture (1)	Face to face	2		
	Theoretical Lab(MS DOS introduction)	Lecture (2)	Face to face	2		
		Lab (2)	Lab	1		
3	System structures; operating system services, user OS interface,	Lecture (1)	Face to face	2		
		Lecture (2)	Face to face	2		
	Getting Started with DOS Commands	Lab (3)	Lab	1		
4	System calls, OS design and implementation	Lecture (1)	Face to face	2		
		Lecture (2)	Face to face	2		
	DOS Network commands	Lab (4)	Lab	1		
5	Process concept; process scheduling, operation on processes,	Lecture (1)	Face to face	2	Quiz1	Summative
		Lecture (2)	Face to face	2		
	Multithreading Using C++ (Part 1)	Lecture (5)	Lab	2		
					Quiz1 Lab	Summative
6	Interprocess communication	Lecture (1)	Face to face	2		
		Lecture (2)	Face to face	2		
	Multithreading Using C++ (Part 1)	Lab (6)	Lab	2		
7	Multithreaded programming; multithreading models, thread libraries, threading issues	Lecture (1)	Face to face	2	Assignment 1	Summative
		Lecture (2)	Face to face	2		
	Midterm Exam (Lab)	Lab (7)	Lab	2	Mid exam Lab	Summative
8	Midterm Exam	Lecture (1)	Face to face	2		
		Lecture (2)	Face to face	2		
		Lab (8)	Lab	2	Mid exam,	Summative

9	Process scheduling; basic concepts, scheduling criteria, scheduling algorithms	Lecture (1)	Face to face	2		
	Process Synchronization (Part 1)	Lecture (2)	Face to face	2		
		Lab (9)	Lab	2		
10	Scheduling algorithms	Lecture (1)	Face to face	2		
	Synchronization; the critical-section problem.	Lecture (2)	Face to face	3		
	Process Synchronization (Part 2)	Lab (10)	Lab	2		
11	Deadlock investigation. Deadlocks; System models, deadlock characterization. Methods for handling deadlocks	Lecture (1)	Face to face	2	Quiz2	Summative
		Lecture (2)	Face to face	2		
		Lab (11)	Lab	2		
					Quiz2 Lab	Summative
12	Memory Management	Lecture (1)	Face to face	3		
	Process Scheduling (Part 1)	Lecture (2)	Face to face	3		
		Lab (12)	Lab	3		
13	Virtual Memory.	Lecture (1)	Face to face	4	Assignment 2	Summative
	Process Scheduling (Part 2)	Lecture (2)	Face to face	4		
		Lab (13)	Lab	2		
14	File System: Tree Structure Directory, Acyclic Graph Structure Directory.	Lecture (1)	Face to face	4		
		Lecture (2)	Face to face	4		
	Final Exam (Lab)	Lab (14)	Lab	1	final exam	Summative

15	File Systems: File System Structure, MBR, Directory Implementation.	Lecture (1)	Face to face	4		
		Lecture (2)	Face to face	4		
16	Final Exam				final exam	Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Written Exam/ Mid Term Exam	25%	Week 8
Written Exam/ Final Exam	40%	Week 16
Quizzes	20%	Week 5, week 11
Assignments	15%	Week 7, week 13

Course Policies

Item	Policy
Quizzes	The format of the quizzes is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes.

	Sign-in sheets will be circulated.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Homework's are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	Lab assignments

Rubrics and Feedbacks (If Applicable)

* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	2	14	28
Tutorials			
laboratory	2	14	28
Activities for Assessment and Evaluation			
First Exam/ Midterm	2	1	2
Second Exam (If applicable)			
Final Exam	4	1	4
Quizzes	½	2	1
Self-Learning Activities			
Labs / Homework	2	8	16
Case Studies			
Presentations			
E-learning			
Extra Readings	3	14	42
Total Hours	121		
JNQF Hours	=121/10 =12.1		